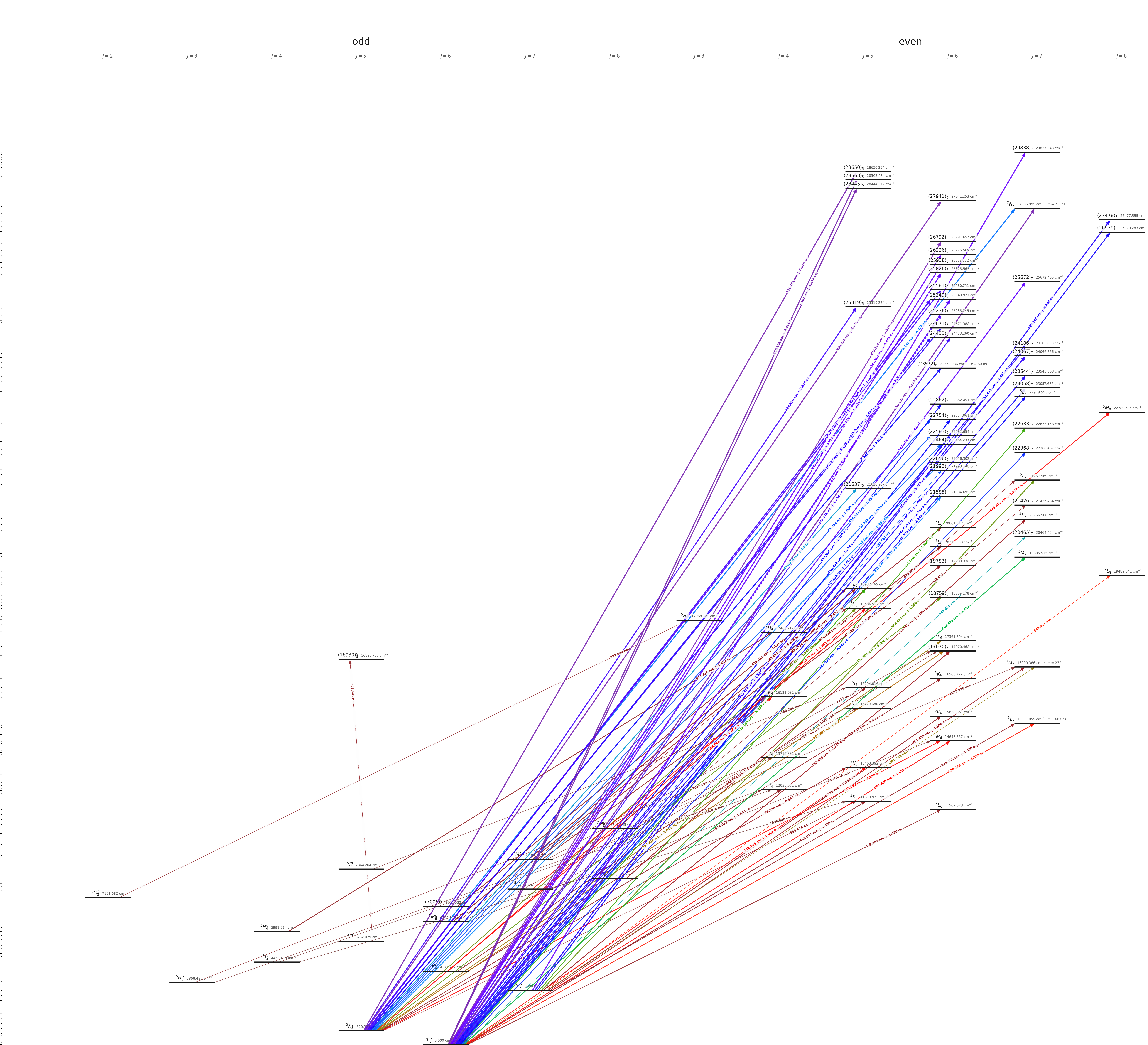


U⁺ ionisation limit 49958.40 cm⁻¹ = 6.19405 eV (200.167 nm)

Arrow label: vacuum wavelength | reduced dipole matrix element $|\langle V \rangle_{er} / \hbar|$ in ea_0
 Matrix elements are measured or NIST values; * = high-accuracy theory, = = model calculation.
 Arrow width grows with the Einstein coefficient A . Dotted: forbidden (clock, M1, E2). Dashed: Rydberg excitation, $n = 70$.
 Level caption: energy (from the cited literature; NIST lists no excited levels) and measured lifetime (where one exists).



Transition	λ vac (nm)	λ air (nm)	ν (THz)	$ J(J+1) $ (ea.)	A (s ⁻¹)	f^{calc} un. (MHz)	branch (%)	Use
¹ _g – (² _g) ₆	396.2633	396.1512	756.5486	–	–	–	–	396.263 nm
¹ _g – ¹ _g	591.7025	591.5385	506.6068	–	–	0.686	–	591.54 nm: classic first-step line of uranium laser spec...
¹ _g – ¹ _g	639.7193	639.5425	468.6312	1.359	9.670e+05	0.2622	58.6	639.542 nm
¹ _g – ¹ _g	713.0865	713.8900	420.4535	1.258	6.800e+05	–	–	713.087 nm
¹ _g – ¹ _g	763.3846	763.1745	392.7140	1.104	3.700e+05	0.686	8.56	763.385 nm
⁷ _g – ¹ _g	788.4112	788.1904	380.2488	3.138	3.700e+06	–	–	788.411 nm: line used for Doppler-free saturated absorpt...
¹ _g – ¹ _g	861.0316	860.7952	348.1782	2.039	1.200e+06	–	–	strongest ground-state absorption line

λ , ν from the level energies of the cited literature (NIST lists no excited levels) unless a measured frequency for this isotope is available. A: measured value or NIST. $\Gamma/2\pi$: natural linewidth of the upper level, $1/(2\pi\tau)$.

N ₂ uphine structure (E = 7/2)										
Level	A (MHz)	B (MHz)	Shift of each F level from the centre of gravity							
¹ N ₂	-60.540	404.010	F=5/2: +3103.57	F=7/2: +2197.25	F=9/2: +1190.70	F=11/2: +202.99	F=13/2: -620.41	F=15/2: -1107.53	F=17/2: -1059.96	F=19/2: -252.84
³ S _g	-68.346	417.054	F=5/2: +1452.64	F=7/2: +1275.57	F=9/2: +1020.91	F=11/2: +714.29	F=13/2: +433.13	F=15/2: -112.36	F=17/2: -619.70	F=19/2: -1186.02
¹ Σ _g ⁺	-56.310	412.000	F=7/12: +3113.03	F=9/2: +2118.90	F=11/2: +1081.36	F=13/2: +106.94	F=15/2: -678.50	F=17/2: -1129.79	F=19/2: -1082.05	F=21/2: -351.60
³ Σ _g ⁺	-59.130	447.000	F=5/2: +1626.47	F=7/2: +1343.32	F=9/2: +966.69	F=11/2: +599.63	F=13/2: +168.12	F=15/2: -278.98	F=17/2: -719.91	F=19/2: -1295.98
¹ Σ _g ⁺	54.890	2366.000	F=5/2: -720.78	F=7/2: -886.07	F=9/2: -1007.31	F=11/2: -1014.66	F=13/2: -823.62	F=15/2: -334.33	F=17/2: +568.45	F=19/2: +2015.30
³ Π _g	67.300	609.000	F=5/2: -1254.47	F=7/2: -1280.60	F=9/2: +926.11	F=11/2: -1529.09	F=13/2: +111.29	F=15/2: -301.58	F=17/2: -679.86	F=19/2: -989.94
¹ Π _g	119.000	619.000	F=9/2: -1723.79	F=11/2: -1157.12	F=13/2: +563.33	F=15/2: -19.71	F=17/2: -548.22	F=19/2: -972.60	F=21/2: -1237.42	
³ Σ _g ⁺	-99.610	956.620	F=5/2: +2484.14	F=7/2: +2090.09	F=9/2: +1568.95	F=11/2: +955.57	F=13/2: -1288.20	F=15/2: -382.21	F=17/2: -1956.51	F=19/2: -1486.45
¹ Σ _g ⁺	20.310	4950.120	F=5/2: +1476.18	F=7/2: +702.83	F=9/2: -98.63	F=11/2: -783.57	F=13/2: -1175.21	F=15/2: -1064.63	F=17/2: -210.79	F=19/2: +1659.45
¹ Σ _g ⁺	-84.720	4422.000	F=7/2: +2042.34	F=9/2: +2846.51	F=11/2: +1597.65	F=13/2: +392.89	F=15/2: -632.99	F=17/2: -1323.61	F=19/2: -1501.94	F=21/2: -970.94
¹ Σ _g ⁺	-171.000	2685.000	F=7/2: +5791.19	F=9/2: +4538.01	F=11/2: +3122.26	F=13/2: +1613.49	F=15/2: -1339.99	F=17/2: -1341.70	F=19/2: -2585.79	F=21/2: -3518.25

Measured A, B constants; sources in the data file

Transition			
λ (nm)	Isotopes	Shift (MHz)	
^4Li - (28563),	350.108 235-238	10340.00	
^6Li - (28445),	351.562 235-238	8990.00	
^7Li -	358.590 235-238	17480.00	
^9Li - (79411),	366.020 235-238	4800.00	
^6Be - (26792),	373.250 235-238	5910.00	
^8Be - (26626),	381.307 235-238	6420.00	
^9Be - (26573),	384.072 235-238	13860.00	
^{10}Be - (25938),	385.531 235-238	8990.00	
^8B - (25876),	387.213 235-238	8180.00	
^{10}B - (25672),	389.522 235-238	10880.00	
^{11}B - (25349),	394.493 235-238	8420.00	
^{12}B - (25236),	396.263 235-238	1260.00	
^{10}C - (25581),	400.634 235-238	4320.00	
^{13}C - (25349),	404.889 235-238	8370.00	
^{12}N - (25319),	404.875 235-238	13280.00	
^{14}N - (24433),	409.278 235-238	5670.00	
^{15}N - (24376),	415.902 235-238	12530.00	
^{16}N - (24067),	415.514 235-238	10300.00	
^{14}O - (24611),	415.780 235-238	5190.00	
^{16}O - (24333),	419.940 235-238	13820.00	
^{17}O - (24748),	422.356 235-238	10460.00	
^{18}O - (23544),	427.456 235-238	8210.00	

$$\text{Shift} = v(\text{first isotope}) - v(\text{second isotope}).$$